

ROUND TUB ARMCHAIR / CNC NOTES

2 sheets 2440 x 1220 x 18 mm (sheet size ASSUMED)

Units / scale	millimetres, 1:1 - never rescale the DXF
Layers	CUT (all closed, cut through) / ENGRAVE-LABEL (0.5-1 mm) / REFERENCE-SHEET (do not cut)
Joints	through-mortises 41 x 19 mm for 40 mm tabs in 18 mm board - UNTESTED FIT
Corner relief	R3.2 dogbone in every inside corner, sized for a 6 mm bit (ASSUMED)
Cutter offset	apply the normal tool-radius offset on every CUT contour; no extra compensation inside slots
Cut order	engrave, then inside contours (mortises, openings), then outlines
Hold-down	parts 10 mm apart, 9 mm clamp edge; add onion-skin or tabs for small ribs
Parts in openings	small ribs are nested inside the ring openings - cut the ribs before the ring's inner contour

[illegible]

Diagram illustrating the arrangement of ribs in a 12-rib cage. The ribs are numbered 1 through 12. Ribs 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, and 12 are shown in a vertical column. Ribs 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, and 12 are shown in a vertical column. Ribs 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, and 12 are shown in a vertical column.

ROUND TUB ARMCHAIR / VALIDATION

automated checks - geometry, joints, 3D assembly

PASS	[1] part topology	01 BASE-RING: 22 loops, simple, holes inside and apart
PASS	[1] part topology	02 SEAT-RING: 35 loops, simple, holes inside and apart
PASS	[1] part topology	03 BODY-RIB-A: 1 loops, simple, holes inside and apart
PASS	[1] part topology	04 BODY-RIB-B: 1 loops, simple, holes inside and apart
PASS	[1] part topology	05 BODY-RIB-C: 1 loops, simple, holes inside and apart
PASS	[1] part topology	06 BODY-RIB-D: 1 loops, simple, holes inside and apart
PASS	[1] part topology	07 BODY-RIB-E: 1 loops, simple, holes inside and apart
PASS	[1] part topology	08 BACK-RIB-A: 1 loops, simple, holes inside and apart
PASS	[1] part topology	09 BACK-RIB-B: 1 loops, simple, holes inside and apart
PASS	[1] part topology	10 BACK-RIB-C: 1 loops, simple, holes inside and apart
PASS	[1] part topology	11 BACK-RIB-D: 1 loops, simple, holes inside and apart
PASS	[1] part topology	12 BACK-RIB-E: 1 loops, simple, holes inside and apart
PASS	[1] part topology	13 BACK-RIB-F: 1 loops, simple, holes inside and apart
PASS	[1] part topology	14 BACK-RIB-G: 1 loops, simple, holes inside and apart
PASS	[1] part topology	15 BACK-BAND-1: 5 loops, simple, holes inside and apart
PASS	[1] part topology	16 BACK-BAND-2: 6 loops, simple, holes inside and apart
PASS	[1] part topology	17 BACK-BAND-3: 5 loops, simple, holes inside and apart
PASS	[2] every tab has its mortise (same angle, same radial span)	base ring: 20/20 body-rib mortises
PASS	[2] every tab has its mortise (same angle, same radial span)	seat ring: 33/33 rib mortises
PASS	[2] every tab has its mortise (same angle, same radial span)	BODY-RIB-A: bottom tab 350.2-390.2 matches its mortise
PASS	[2] every tab has its mortise (same angle, same radial span)	BODY-RIB-B: bottom tab 367.0-407.0 matches its mortise
PASS	[2] every tab has its mortise (same angle, same radial span)	BODY-RIB-C: bottom tab 373.3-413.3 matches its mortise
PASS	[2] every tab has its mortise (same angle, same radial span)	BODY-RIB-D: bottom tab 357.4-397.4 matches its mortise
PASS	[2] every tab has its mortise (same angle, same radial span)	BODY-RIB-E: bottom tab 337.4-377.4 matches its mortise
PASS	[2] every tab has its mortise (same angle, same radial span)	back bands carry 13 rib slots for 13 back ribs
PASS	[2] every tab has its mortise (same angle, same radial span)	every back rib rises at least 20 mm above the band
PASS	[3] slot sizes (Tokyo rule: 19 x 41 for 18 mm board)	mortise 41 x 19 mm for 18 mm board
PASS	[3] slot sizes (Tokyo rule: 19 x 41 for 18 mm board)	relief R3.2 clears a 6 mm cutter
PASS	[4] corner relief and cutter reach	01 BASE-RING: 0 corners relieved, worst spot a 6 mm bit cannot reach 0.00 mm2
PASS	[4] corner relief and cutter reach	02 SEAT-RING: 0 corners relieved, worst spot a 6 mm bit cannot reach 0.00 mm2
PASS	[4] corner relief and cutter reach	03 BODY-RIB-A: 4 corners relieved, worst spot a 6 mm bit cannot reach 0.00 mm2
PASS	[4] corner relief and cutter reach	04 BODY-RIB-B: 4 corners relieved, worst spot a 6 mm bit cannot reach 0.00 mm2
PASS	[4] corner relief and cutter reach	05 BODY-RIB-C: 4 corners relieved, worst spot a 6 mm bit cannot reach 0.00 mm2
PASS	[4] corner relief and cutter reach	06 BODY-RIB-D: 4 corners relieved, worst spot a 6 mm bit cannot reach 0.00 mm2
PASS	[4] corner relief and cutter reach	07 BODY-RIB-E: 4 corners relieved, worst spot a 6 mm bit cannot reach 0.00 mm2
PASS	[4] corner relief and cutter reach	08 BACK-RIB-A: 4 corners relieved, worst spot a 6 mm bit cannot reach 0.00 mm2
PASS	[4] corner relief and cutter reach	09 BACK-RIB-B: 4 corners relieved, worst spot a 6 mm bit cannot reach 0.00 mm2
PASS	[4] corner relief and cutter reach	10 BACK-RIB-C: 4 corners relieved, worst spot a 6 mm bit cannot reach 0.00 mm2
PASS	[4] corner relief and cutter reach	11 BACK-RIB-D: 4 corners relieved, worst spot a 6 mm bit cannot reach 0.00 mm2
PASS	[4] corner relief and cutter reach	12 BACK-RIB-E: 4 corners relieved, worst spot a 6 mm bit cannot reach 0.00 mm2
PASS	[4] corner relief and cutter reach	13 BACK-RIB-F: 4 corners relieved, worst spot a 6 mm bit cannot reach 0.00 mm2
PASS	[4] corner relief and cutter reach	14 BACK-RIB-G: 4 corners relieved, worst spot a 6 mm bit cannot reach 0.00 mm2
PASS	[4] corner relief and cutter reach	15 BACK-BAND-1: 0 corners relieved, worst spot a 6 mm bit cannot reach 0.00 mm2
PASS	[4] corner relief and cutter reach	16 BACK-BAND-2: 0 corners relieved, worst spot a 6 mm bit cannot reach 0.00 mm2
PASS	[4] corner relief and cutter reach	17 BACK-BAND-3: 0 corners relieved, worst spot a 6 mm bit cannot reach 0.00 mm2
PASS	[5] nesting	sheet 1: 28 parts, closest 12.9 mm apart, all inside the 9 mm edge
PASS	[5] nesting	sheet 2: 10 parts, closest 13.3 mm apart, all inside the 9 mm edge
PASS	[5] nesting	every part placed its full quantity (38 pieces)
PASS	[6] labels	38 pieces labelled
PASS	[6] labels	every label sits on material, 4 mm from any cut

ROUND TUB ARMCHAIR / VALIDATION (cont.)

- PASS [7] built DXF units = mm
- PASS [7] built DXF 106 closed CUT profiles (expected 106)
- PASS [7] built DXF 38 ENGRAVE-LABEL texts
- PASS [8] DXF hygiene, scale and cutter compatibility no duplicate CUT profiles (0 found)
- PASS [8] DXF hygiene, scale and cutter compatibility scale 1:1 (base ring 817.0 mm in model and in the DXF)
- PASS [8] DXF hygiene, scale and cutter compatibility narrowest internal feature 19 mm > 6 mm cutter + 1 mm
- PASS [8] DXF hygiene, scale and cutter compatibility thinnest ring web beside a mortise 16.5 mm (>= 12 mm)
- PASS [8] DXF hygiene, scale and cutter compatibility mirrored rib positions share one profile (plan symmetric in X and Y)
- PASS [3D-1] interference 88 neighbouring pairs, 0 interpenetrate
- PASS [3D-2] load path: every rib bears on what carries it 20/20 body ribs touch both rings
- PASS [3D-2] load path: every rib bears on what carries it 13/13 back ribs touch the seat ring and a band
- PASS [3D-2] load path: every rib bears on what carries it 3/3 bands rest on back ribs
- PASS [3D-3] overall size frame 953.5 x 922.8 x 800.0 mm, standing on z = 0